

Reproducing Deep Learning Skin Lesion Classification on HAM10000

Comparing Transfer Learning Architectures

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- Skin cancer is the most common cancer type globally
- Early detection: 5-year melanoma survival $>95\%$ (early) vs $<30\%$ (late)
- Dermoscopy accuracy depends on clinician experience
- Deep learning has matched dermatologist-level accuracy (Esteva et al., 2017)

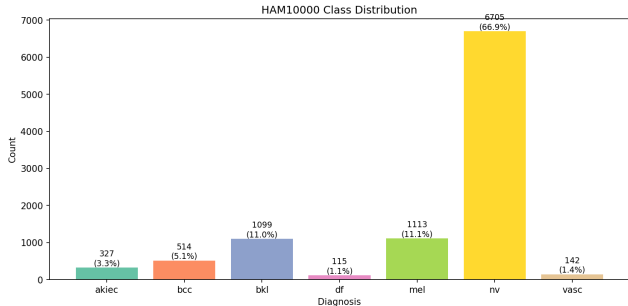
This project: reproduce the transfer learning approach on HAM10000

- Compare two architectures: ResNet-50 vs EfficientNet-B0
- Use Grad-CAM to visualise what the models focus on

Dataset: HAM10000

- 10,015 dermoscopic images
- 7 diagnostic categories
- Severe class imbalance (67% nv)
- Split by lesion_id (not image_id) to prevent data leakage

Tschandl et al. (2018), Scientific Data



Transfer learning:

- Both models pretrained on ImageNet (1.2M natural images)
- Freeze early layers, fine-tune last block + classifier head
- ResNet-50: ~15M trainable params / EfficientNet-B0: ~1.1M trainable params

Handling class imbalance:

- Weighted cross-entropy loss (inverse frequency weighting)
- Data augmentation: random flips, rotation, colour jitter

Training:

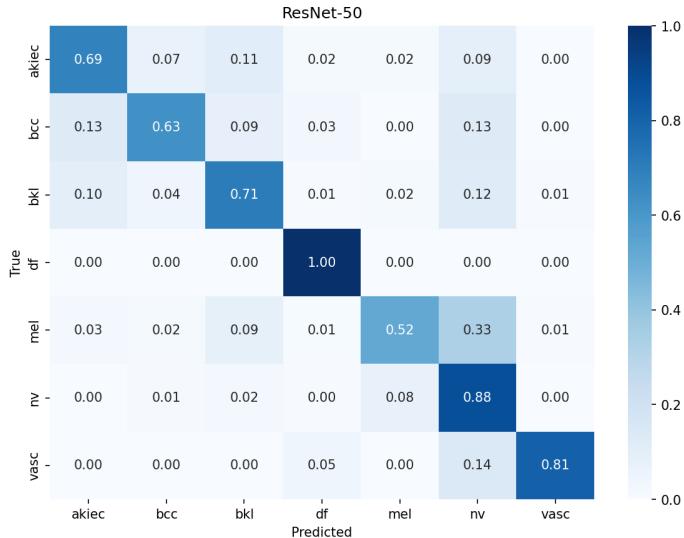
- Adam optimiser, $LR = 1e-4$, batch size 32, images resized to 224x224
- Early stopping (patience = 5 epochs on validation loss)

Results

Model	Accuracy	Macro F1	Weighted F1
ResNet-50	79.7%	0.674	0.799
EfficientNet-B0	71.2%	0.544	0.734

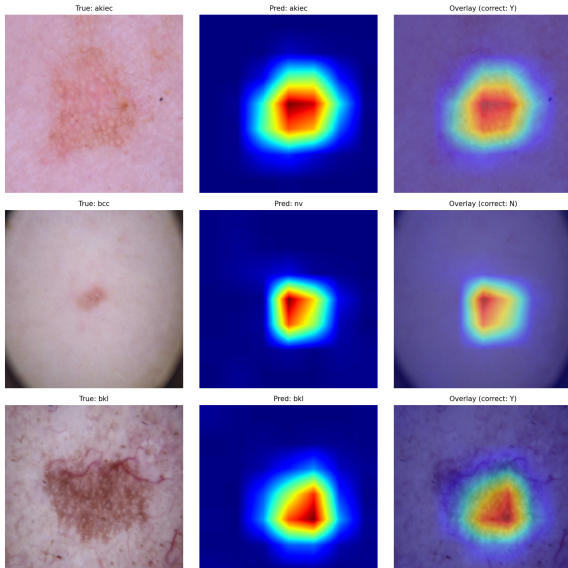
- ResNet-50 clearly outperforms EfficientNet-B0 in this setup
- Hardest class: melanoma (52% recall) - confused with melanocytic nevi
- Easiest class: melanocytic nevi (88% recall) - most training data
- EfficientNet gap likely due to too few trainable parameters

Results - Confusion Matrix (ResNet-50)



Grad-CAM Analysis

Grad-CAM Attention Maps - resnet50



- Models focus on the lesion itself (not artifacts or borders)
- ResNet-50 produces more focused heatmaps than EfficientNet-B0
- Melanoma: attention on colour variation and irregular borders
- Suggests clinically relevant features learned

Columns: original image, heatmap, overlay. Grad-CAM run on both models (ResNet-50 shown).

Summary:

- Reproduced transfer learning classification on HAM10000
- ResNet-50 (79.7%) outperformed EfficientNet-B0 (71.2%)
- Grad-CAM confirms models attend to clinically relevant features

Limitations:

- Single-image classification (no temporal info, no patient metadata)
- Small dataset from only two institutions
- EfficientNet may improve with more unfrozen layers

Code: github.com/daniel-duhnev/advanced-topics-upf